

Sisteme de ec. diferentiale

$$1) \begin{cases} x' = 4x - 3y \\ y' = 3x + 4y \end{cases}$$

$$\begin{cases} x' = 4x - 3y \\ y' = 3x + 4y \end{cases}$$

$$y = \frac{1}{3}(4x - x')$$

$$y' = \frac{1}{3}(4x' - x'')$$

$$\frac{1}{3}(4x' - x'') = 3x + 4 \cdot \frac{1}{3}(4x - x') \quad | \cdot 3$$

$$4x' - x'' = 9x + 16x - 4x'$$

$$x'' - 8x' + 25x = 0$$

Ec. caract : $n^2 - 8n + 25 = 0$

$$\Delta = 64 - 100 = -36$$

$$n_{1,2} = \frac{8 \pm i6}{2} \Rightarrow \begin{cases} n_1 = 4 + 3i \\ n_2 = 4 - 3i \end{cases}$$

$$y_1(t) = e^{4t} \cdot \cos(3t)$$

$$y_2(t) = e^{4t} \sin(3t)$$

$$x(t) = C_1 \cdot e^{4t} \cos(3t) + C_2 \cdot e^{4t} \sin(3t)$$

$$x'(t) = C_1 \cdot 4 e^{4t} \cdot \cos(3t) - C_1 \cdot e^{4t} \cdot \sin(3t) \cdot 3 \\ + 4 C_2 e^{4t} \sin(3t) + 3 C_2 e^{4t} \cos(3t)$$

$$y(t) = \frac{1}{3} (4x(t) - x'(t))$$

$$= \frac{1}{3} \left(\cancel{4 C_1 e^{4t} \cos(3t)} + \cancel{4 C_2 e^{4t} \sin(3t)} - \right. \\ \left. - \cancel{C_1 \cdot 4 e^{4t} \cos(3t)} + C_1 e^{4t} \cdot \sin(3t) \cdot 3 - \cancel{4 C_2 e^{4t} \sin(3t)} \right. \\ \left. - 3 C_2 e^{4t} \cos(3t) \right) \\ = C_1 e^{4t} \sin(3t) - C_2 e^{4t} \cos(3t)$$

2) a)
$$\begin{cases} x' = -2x - 4y + 1 + 4t \\ y' = -x + y + \frac{3}{2}t^2 \end{cases}$$

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$$y' = -x + y + \frac{3}{2} t^2$$

$$y = \frac{1}{4}(-2x - x' + 1 + 4t)$$

$$y' = \frac{1}{4}(-2x' - x'')$$

$$\frac{1}{4}(-2x' - x'') = -x + \frac{1}{4}(-2x - x' + 1 + 4t) + \frac{3}{2} t^2 \quad | \cdot 4$$

$$-2x' - x'' = -4x - 2x - x' + 1 + 4t + 6t^2$$

$$x'' + x' - 6x = 6t^2 + 4t + 1 = 0$$

$$x'' + x' - 6x = 6t^2 + 4t + 1$$

Ec. caract : $r^2 + r - 6 = 0$

$$\Delta = 1 + 24 = 25$$

$$r_{1,2} = \frac{-1 \pm 5}{2} \Rightarrow \begin{cases} r_1 = 2 \\ r_2 = -3 \end{cases}$$

$$x_1(t) = e^{2t}$$

$$x_2(t) = e^{-3t}$$

$$\Rightarrow x_0(t) = e^{2t} C_1 + e^{-3t} C_2$$

$$f(t) = -6t^2 - 4t - 1$$

$\alpha = 0 \Rightarrow e^0 = 1 \rightarrow$ nu e rădăcină

$$P_m(t) = -6t^2 - 4t - 1 \rightarrow \text{grad } \underline{u} \Rightarrow Q(t) = at^2 + bt + c$$

$\rightarrow$ verifică ec.

$$x'' + x' - 6x = -6t^2 - 4t - 1$$

$$\Rightarrow 2a + 2at + b - 6at^2 - 6bt - 6c = -6t^2 - 4t - 1$$

$$-t^2 \cdot 6a - t(-2a + 6b) - (-2a - b + 6c) = -6t^2 - 4t - 1$$

$$\Rightarrow \begin{cases} 6a = 6 \Rightarrow a = 1 \end{cases}$$

$$\begin{cases} -2a + 6b = 4 \Rightarrow b = \frac{4 + 2 \cdot 1}{6} = \frac{6}{6} = 1 \end{cases}$$

$$\begin{cases} -2a - b + 6c = -1 \Rightarrow c = \frac{1 + 2 \cdot 1 + 1}{6} = \frac{4}{6} = \frac{2}{3} \end{cases}$$

$$\Rightarrow x_p(t) = t^2 + t + \frac{2}{3}$$

$$\Rightarrow x(t) = e^{2t} C_1 + e^{-3t} C_2 + t^2 + t + \frac{2}{3}$$

$$x'(t) = 2e^{2t} C_1 - 3e^{-3t} C_2 + 2t + 1$$

$$y(t) = \frac{1}{5} (-2x(t) - x'(t) + 1 + 4t)$$

pour 5 (c'est tout)

$$= \frac{1}{5} \left(-2e^{2t} C_1 - 2e^{-3t} C_2 - 2t^2 - \cancel{2t} - \frac{4}{3} - \right. \\ \left. - 2e^{2t} C_1 + 3e^{-3t} C_2 - \cancel{2t} - \cancel{1} + \cancel{1} + \cancel{1} \right) \\ = \frac{1}{5} \left(-4e^{2t} C_1 + e^{-3t} C_2 - 2t^2 - \frac{4}{3} \right)$$

3) 3) $\begin{cases} x' = x - y + 3t^2 \\ y' = -4x - 2y + 8t + 2 \end{cases}$

$$\begin{cases} x' = x - y + 3t^2 \\ y' = -4x - 2y + 8t + 2 \end{cases}$$

$$y = x - x' + 3t^2$$

$$y' = -x'' + x'$$

$$\Rightarrow -x'' + x' = -4x - 2x + 2x' - 6t^2 + 8t + 2$$

$$x'' + x' - 6x = 6t^2 - 8t - 2$$

Ec. caract: $r^2 + r - 6 = 0$

$$\Delta = 1 + 24 = 25$$

$$r_{1,2} = \frac{-1 \pm 5}{2} \Rightarrow \begin{cases} r_1 = \frac{4}{2} = 2 \\ r_2 = -\frac{6}{2} = -3 \end{cases}$$

$$\Rightarrow x_1(t) = e^{2t}$$

$$x_2(t) = e^{-3t}$$

$$\Rightarrow x_h(t) = e^{2t} C_1 + e^{-3t} C_2$$

$$f(t) = 6t^2 - 8t - 2$$

$\alpha = 0 \rightarrow$ nu e r dăcină

$$P_m(t) = 6t^2 - 8t - 2 \rightarrow \text{grad } \underline{u} \Rightarrow Q_m(t) = at^2 + bt + c$$

$$\Rightarrow x'' + x' - 6x = 6t^2 - 8t - 2$$

$$\Rightarrow 2a + 2at + b - 6at^2 - 6bt - 6c = 6t^2 - 8t - 2$$

$$t^2 \cdot (-6a) - t(-2a + 6b) - (-2a - b + 6c) = 6t^2 - 8t - 2$$

$$\Rightarrow \begin{cases} -6a = 6 \Rightarrow a = -1 \\ -2a + 6b = 8 \Rightarrow b = \frac{8 + 2 \cdot (-1)}{6} = \frac{6}{6} = 1 \\ -2a - b + 6c = 2 \Rightarrow c = \frac{2 + 2 \cdot (-1) + 1}{6} = \frac{1}{6} \end{cases}$$

$$\Rightarrow x_p(t) = -t^2 + t + \frac{1}{6}$$

$$\Rightarrow x(t) = e^{2t} C_1 + e^{-3t} C_2 - t^2 + t + \frac{1}{6}$$

$$x'(t) = 2e^{2t} C_1 - 3e^{-3t} C_2 - 2t + 1$$

$$x'(t) = 2e^{2t} C_1 - 3e^{-3t} C_2 - 2t + 1$$

$$y(t) = -x(t)' + x(t) + 3t^2$$

$$y(t) = -2e^{2t} C_1 + 3e^{-3t} C_2 + 2t - 1 + e^{2t} C_1 + e^{-3t} C_2 - t^2 + t + \frac{1}{6} + 3t^2$$

$$= -e^{2t} C_1 + 4e^{-3t} C_2 + 2t^2 + 3t - \frac{5}{6}$$

$$4) \begin{cases} x' = y + t \\ y' = x + e^{-t} \end{cases}$$

$$\begin{cases} x' = y + t \\ y' = x + e^{-t} \end{cases}$$

$$y = x' - t$$

$$y' = x''$$

$$x'' = x + e^{-t}$$

$$x'' - x = e^{-t}$$

$$\text{Ec. caract : } r^2 - 1 = 0 \Leftrightarrow (r-1)(r+1) = 0 \Rightarrow \begin{cases} r_1 = 1 \\ r_2 = -1 \end{cases}$$

$$\Rightarrow x_1(t) = e^t$$

$$x_2(t) = e^{-t}$$

$$\Rightarrow x_h(t) = e^t C_1 + e^{-t} C_2$$

$$f(t) = e^{-t}$$

$$\Rightarrow \alpha = -1 \Rightarrow \text{röd. de ord } \Sigma$$

$$P_m(t) = 1$$

$$\Rightarrow x_p(t) = e^{-t} \cdot t \cdot a$$

$$x'_p(t) = -e^{-t} \cdot t \cdot a + e^{-t} \cdot a$$

$$\begin{aligned} x''_p(t) &= e^{-t} \cdot t \cdot a - e^{-t} \cdot a - e^{-t} \cdot a \\ &= e^{-t} \cdot t \cdot a - 2 \cdot e^{-t} \cdot a \end{aligned}$$

$$x'' - x = e^{-t}$$

$$\cancel{e^{-t} \cdot t \cdot a} - 2 \cdot e^{-t} \cdot a - \cancel{e^{-t} \cdot t \cdot a} = e^{-t}$$

$$-2 \cancel{e^{-t}} \cdot a = \cancel{e^{-t}} \Rightarrow -2a = 1 \Rightarrow a = -\frac{1}{2}$$

$$\Rightarrow x_p(t) = -\frac{e^{-t} \cdot t}{2}$$

$$\Rightarrow x(t) = e^t C_1 + e^{-t} C_2 - \frac{e^{-t} \cdot t}{2}$$

$$x'(t) = e^t C_1 - e^{-t} C_2 - \frac{1}{2} (-e^{-t} \cdot t + e^{-t})$$

$$x'(t) = e^t C_1 - e^{-t} C_2 - \frac{1}{2}(-e^{-t} \cdot t + e^{-t})$$

$$y = x' - t$$

$$y(t) = x(t)' - t$$

$$y(t) = e^t C_1 - e^{-t} C_2 - \frac{1}{2} \cdot e^{-t} (1 - t) - t$$

5) 5) $\begin{cases} x' = -x + y + z \\ y' = x - y + z \\ z' = x + y + z \end{cases}$

$$\begin{cases} x' = -x + y + z \\ y' = x - y + z \\ z' = x + y + z \end{cases} \Rightarrow A = \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -1-\lambda & 1 & 1 \\ 1 & -1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix}$$

$$= (1+\lambda)^2 \cdot (1-\lambda) + 1 + 1 + 1 + \lambda \quad \cancel{1+\lambda} \quad \cancel{1+\lambda}$$

$$= (1+\lambda)^2 (1-\lambda) + 3(1+\lambda) = (1+\lambda) ((1+\lambda)(1-\lambda) + 3)$$

$$\Rightarrow 1 + \lambda = 0 \Rightarrow \lambda_1 = -1$$

$$1 - \lambda^2 + 3 = 0 \Rightarrow \lambda^2 = 4 \Rightarrow \left. \begin{matrix} \lambda_2 = 2 \\ \lambda_3 = -2 \end{matrix} \right\}$$

$$1 - \lambda^2 + 3 = 0 \Rightarrow \lambda^2 = 4 \Rightarrow \begin{cases} \lambda_2 = 2 \\ \lambda_3 = -2 \end{cases}$$

$$\underline{I.} \lambda_1 = 1$$

$$\begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \lambda_1 \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

$$\begin{cases} -a_1 + a_2 + a_3 = a_1 \\ a_1 - a_2 + a_3 = a_2 \\ a_1 + a_2 + a_3 = a_3 \end{cases} \Rightarrow \begin{cases} -2a_1 + a_2 + a_3 = 0 \\ a_1 - 2a_2 + a_3 = 0 \\ a_1 + a_2 = 0 \end{cases}$$

$$B = \begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} = 4 - 1 = 3 \neq 0$$

$$\Rightarrow \text{not: } a_3 = \alpha$$

$$\Rightarrow \begin{cases} -2a_1 + a_2 = -\alpha \\ a_1 - 2a_2 = -\alpha \end{cases} \quad \begin{array}{l} | \cdot 2 \\ \oplus \end{array}$$

$$-3a_1 = -3\alpha \Rightarrow a_1 = \alpha$$

$$a_2 = -\alpha + 2a_1 = -\alpha + 2\alpha = \alpha$$

$$\Rightarrow \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\underline{I.} \lambda_2 = 2$$

$$\begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \pi_2 \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$\Rightarrow \begin{cases} -b_1 + b_2 + b_3 = 2b_1 \\ b_1 - b_2 + b_3 = 2b_2 \\ b_1 + b_2 + b_3 = 2b_3 \end{cases} \Rightarrow \begin{cases} -3b_1 + b_2 + b_3 = 0 \\ b_1 - 3b_2 + b_3 = 0 \\ b_1 + b_2 - b_3 = 0 \end{cases}$$

$$\beta = \begin{vmatrix} -3 & 1 \\ 1 & -3 \end{vmatrix} = 9 - 1 = 8 \neq 0$$

$$\Rightarrow \text{met: } 3 = \beta$$

$$\begin{cases} -3b_1 + b_2 = -\beta & | \cdot 3 \\ b_1 - 3b_2 = -\beta & \oplus \end{cases}$$

$$-8b_1 = -4\beta \Rightarrow b_1 = \frac{\beta}{2}$$

$$b_2 = \frac{2}{-1}\beta + 3 \cdot \frac{\beta}{2} = \frac{\beta}{2}$$

$$\Rightarrow \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \beta \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{pmatrix}$$

$$\underline{\overline{m}} \cdot \pi_3 = -2$$

$$\Rightarrow \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = R_3 \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$\begin{cases} -c_1 + c_2 + c_3 = -2c_1 \\ c_1 - c_2 + c_3 = -2c_2 \\ c_1 + c_2 + c_3 = -2c_3 \end{cases} \Rightarrow \begin{cases} c_1 + c_2 + c_3 = 0 \\ c_1 + c_2 + c_3 = 0 \\ c_1 + c_2 + 3c_3 = 0 \end{cases}$$

$$b_2 \begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} = 3 - 1 = 2 \neq 0$$

$$\Rightarrow \text{mat: } c_1 = \gamma$$

$$\Rightarrow \begin{cases} c_2 + c_3 = -\gamma & | \cdot (-3) \\ c_2 + 3c_3 = -\gamma \end{cases}$$

$$\begin{array}{r} \hline -2c_2 = 2\gamma \rightarrow c_2 = -\gamma \end{array} \quad \textcircled{+}$$

$$c_3 = -\gamma - c_2 = 0$$

$$\Rightarrow \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \gamma \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = C_1 \cdot e^t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + C_2 \cdot e^{\frac{1}{2}t} \begin{pmatrix} \frac{1}{2} \\ 1 \\ 1 \end{pmatrix} + C_3 \cdot e^{-2t} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

6) $\begin{cases} x' = -2x - 3y - 2z \\ y' = -5x - 4y + 2z \\ z' = 3x + y + 3z \end{cases}$

$$\begin{cases} x' = -2x - 3y - 2z \\ y' = -5x - 4y + 2z \\ z' = 3x + y + 3z \end{cases} \Rightarrow A = \begin{pmatrix} -2 & -3 & -2 \\ -5 & -4 & 2 \\ 3 & 1 & 3 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -2-\lambda & -3 & -2 \\ -5 & -4-\lambda & 2 \\ 3 & 1 & 3-\lambda \end{vmatrix} =$$

$$= (2+\lambda)(4+\lambda)(3-\lambda) - 18 + 10 - 24 - 6\lambda - 45 + 15\lambda + 4 + 2\lambda$$

$$= (2+\lambda)(4+\lambda)(3-\lambda) + 11\lambda - 73$$

$$= -\lambda^3 - 3\lambda^2 + 21\lambda - 49$$

$$= (\lambda+7)(\lambda^2 - 4\lambda + 7)$$

$$\Rightarrow \lambda + 7 = 0 \Rightarrow \lambda_1 = -7$$

$$\lambda^2 - 4\lambda + 7 = 0$$

$$\Delta = 16 - 28 = -12$$

$$\lambda_{2,3} = \frac{4 \pm 2\sqrt{3}i}{2} \Rightarrow \begin{cases} \lambda_2 = 2 + \sqrt{3}i \\ \lambda_3 = 2 - \sqrt{3}i \end{cases}$$

$$\underline{\Sigma. \lambda_1 = -7}$$

$$\begin{pmatrix} -2 & -3 & -2 \\ -5 & -4 & 2 \\ 3 & 1 & 3 \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \lambda_1 \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

$$\begin{cases} -2a_1 - 3a_2 - 2a_3 = -7a_1 \\ -5a_1 - 4a_2 + 2a_3 = -7a_2 \\ 3a_1 + a_2 + 3a_3 = -7a_3 \end{cases} \Rightarrow \begin{cases} 5a_1 - 3a_2 - 2a_3 = 0 \\ -5a_1 + 3a_2 + 2a_3 = 0 \\ 3a_1 + a_2 + 10a_3 = 0 \end{cases}$$

$$B = \begin{vmatrix} -5 & 3 \\ 3 & 1 \end{vmatrix} = -5 - 9 = -14 \neq 0$$

$$\Rightarrow \text{not: } a_3 = \alpha$$

$$\Rightarrow \begin{cases} -5a_1 + 3a_2 = -2\alpha \\ 3a_1 + a_2 = -10\alpha \end{cases} \quad | \cdot (-3) \oplus$$

$$-14a_1 = 28\alpha \Rightarrow a_1 = -2\alpha$$

$$a_2 = -10\alpha - 3 \cdot (-2\alpha) = -10\alpha + 6\alpha = -4\alpha$$

$$\Rightarrow \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \alpha \begin{pmatrix} -2 \\ -4 \\ 1 \end{pmatrix}$$

$$\text{ii. } \pi_2 = 2 + \sqrt{3}i$$

$$\begin{pmatrix} -2 & -3 & -2 \\ -5 & -4 & 2 \\ 3 & 1 & 3 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \pi_2 \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$\begin{cases} -2b_1 - 3b_2 - 2b_3 = 2b_1 + \sqrt{3}i b_1 \\ -5b_1 - 4b_2 + 2b_3 = 2b_2 + \sqrt{3}i b_2 \\ 3b_1 + b_2 + 3b_3 = 2b_3 + \sqrt{3}i b_3 \end{cases}$$

$$\begin{cases} b_1(-4 - \sqrt{3}i) - 3b_2 - 2b_3 = 0 \\ -5b_1 + b_2(-6 - \sqrt{3}i) + 2b_3 = 0 \\ 3b_1 + b_2 + b_3(1 - \sqrt{3}i) = 0 \end{cases}$$

$$\beta = \begin{vmatrix} -4 - \sqrt{3}i & -3 \\ -5 & -6 - \sqrt{3}i \end{vmatrix} = 24 + 4\sqrt{3}i + 6\sqrt{3}i - 3 - 15 = 10\sqrt{3}i + 6 \neq 0$$

$$\Rightarrow \text{not: } b_3 = \beta$$

$$\Rightarrow \text{mat: } b_3 = \beta$$

$$\begin{cases} b_1(-4 - \sqrt{3}i) - 3b_2 = 2\beta & |(-6 + \sqrt{3}i)| \\ -5b_1 + b_2(-8 - \sqrt{3}i) = -2\beta & | \cdot 3 \end{cases} \quad (+)$$

$$\left(\begin{pmatrix} 24 + 4\sqrt{3}i + 8\sqrt{3}i - 15 \end{pmatrix} - 15 \right) b_1 - (-12 - 2\sqrt{3}i - 6)\beta$$

$$(10\sqrt{3}i + 9) \cdot b_1 = (-18 - 2\sqrt{3}i)\beta$$

$$b_1 = \frac{-18 - 2\sqrt{3}i}{6 + 10\sqrt{3}i} \beta = \frac{-1 + \sqrt{3}i}{2} \beta$$

$$b_2 = \frac{\frac{2}{2}\beta + \frac{(4 + \sqrt{3}i) \cdot (-1 + \sqrt{3}i)}{2}\beta}{-3} = \frac{\cancel{4} - 4 + 4\sqrt{3}i - \sqrt{3}i - 3}{-6}$$

$$= \frac{-3 + 3\sqrt{3}i}{-6} = \frac{1 - \sqrt{3}i}{2}$$

$$\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \beta \begin{pmatrix} \frac{-1 + \sqrt{3}i}{2} \\ \frac{1 - \sqrt{3}i}{2} \\ 1 \end{pmatrix}$$

$$\text{III. } \lambda_3 = 2 - \sqrt{3}i$$

$$\begin{pmatrix} -2 & -3 & -2 \\ -5 & -4 & 2 \\ 3 & 1 & 3 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \lambda_3 \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$\begin{cases} -2c_1 - 3c_2 - 2c_3 = (2 - \sqrt{3}i)c_1 \\ -5c_1 - 4c_2 + 2c_3 = (2 - \sqrt{3}i)c_2 \\ 3c_1 + c_2 + 3c_3 = (2 - \sqrt{3}i)c_3 \end{cases}$$

$$\Rightarrow \begin{cases} c_1(-4 + \sqrt{3}i) - 3c_2 - 2c_3 = 0 \\ -5c_1 + c_2(-6 + \sqrt{3}i) + 2c_3 = 0 \\ 3c_1 + c_2 + c_3(1 + \sqrt{3}i) = 0 \end{cases}$$

$$B = \begin{vmatrix} -4 + \sqrt{3}i & -3 \\ -5 & -6 + \sqrt{3}i \end{vmatrix} = 24 - 4\sqrt{3}i - 6\sqrt{3}i - 3 - 15 \\ = -10\sqrt{3}i + 6 \neq 0$$

$$\Rightarrow \text{mat}, c_3 = \varphi$$

$$\begin{cases} (-4 + \sqrt{3}i)c_1 - 3c_2 = 2\varphi & | \cdot (-6 + \sqrt{3}i) \\ -5c_1 + (-6 + \sqrt{3}i)c_2 = -2\varphi & | \cdot 3 \end{cases} \quad (+)$$

$$(-10\sqrt{3}i + 6)c_1 = \varphi(-12 + 2\sqrt{3}i - 6)$$

$$c_1 = \frac{-18 + 2\sqrt{3}i}{6 - 10\sqrt{3}i} \varphi = \frac{-1 - \sqrt{3}i}{2} \varphi$$

$$c_2 = \frac{\cancel{2}^2 \varphi + \frac{(1 + \sqrt{3}i)(-4 + \sqrt{3}i)}{2} \varphi}{-3} = \frac{(\cancel{4} - 4 + \sqrt{3}i - 4\sqrt{3}i - 3)\varphi}{-6} \\ = \frac{-3 - 3\sqrt{3}i}{-6} = \frac{1 + \sqrt{3}i}{2}$$

$$\Rightarrow \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = \varphi \begin{pmatrix} \frac{-1 - \sqrt{3}i}{2} \\ \frac{1 + \sqrt{3}i}{2} \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = C_1 e^{2t} \begin{pmatrix} -2 \\ -4 \\ 1 \end{pmatrix} + C_2 \cdot e^{2t} \cos(\sqrt{3}) \cdot \begin{pmatrix} \frac{-1 + \sqrt{3}i}{2} \\ \frac{1 - \sqrt{3}i}{2} \\ 1 \end{pmatrix} + C_3 \cdot e^{2t} \sin(\sqrt{3}) \cdot \begin{pmatrix} \frac{-1 - \sqrt{3}i}{2} \\ \frac{1 + \sqrt{3}i}{2} \\ 1 \end{pmatrix}$$

$$7) \Rightarrow \begin{cases} x' = x - y - z \\ y' = 3x + y - 3z \\ z' = -4x - 2y + z \end{cases}$$

$$\begin{cases} x' = x - y - z \\ y' = 3x + y - 3z \\ z' = -4x - 2y + z \end{cases} \Rightarrow A = \begin{pmatrix} 1 & -1 & -1 \\ 3 & 1 & -3 \\ -4 & -2 & 1 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -1 & -1 \\ 3 & 1-\lambda & -3 \\ -4 & -2 & 1-\lambda \end{vmatrix}$$

$$= (1-\lambda)^3 - 12 + 6 - 4 + 4\lambda + 3 - 3\lambda - 6 + 6\lambda$$

$$= (1 - \lambda)^3 + 7\lambda - 13$$

$$= -\lambda^3 + 3\lambda^2 + 4\lambda - 12$$

$$\Rightarrow (3 - \lambda)(4 - \lambda^2) = 0$$

$$\Rightarrow \begin{cases} \lambda_1 = 3 \\ \lambda^2 = 4 \end{cases} \Rightarrow \begin{cases} \lambda_2 = 2 \\ \lambda_3 = -2 \end{cases}$$

$$\underline{1. \lambda_1 = 3}$$

$$\begin{pmatrix} 1 & -1 & -1 \\ 3 & 1 & -3 \\ -4 & -2 & 1 \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \lambda_1 \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

$$\begin{cases} a_1 - a_2 - a_3 = 3a_1 \\ 3a_1 + a_2 - 3a_3 = 3a_2 \\ -4a_1 - 2a_2 + a_3 = 3a_3 \end{cases} \Rightarrow \begin{cases} -2a_1 - a_2 - a_3 = 0 \\ 3a_1 - 2a_2 - 3a_3 = 0 \\ -4a_1 - 2a_2 - 2a_3 = 0 \end{cases}$$

$$b = \begin{vmatrix} -2 & -1 \\ 3 & -1 \end{vmatrix} = 2 + 3 = 5 \neq 0$$

$$\Rightarrow \text{not: } a_3 = \alpha$$

$$\begin{cases} -2a_1 - a_2 = \alpha & | \cdot (-2) \\ 3a_1 - 2a_2 = 3\alpha & \oplus \end{cases}$$

$$4a_1 - 2a_2 = -2\alpha \quad a_1 = \frac{\alpha}{-2}$$

$$\frac{1 \cdot a_1 - 1 \cdot a_2 = 0 \alpha \quad (+)}{7a_1 = \alpha \Rightarrow a_1 = \frac{\alpha}{7}}$$

$$\Rightarrow a_2 = -2 \cdot \frac{\alpha}{7} - \frac{\alpha}{7} = -\frac{3\alpha}{7}$$

$$\Rightarrow \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \alpha \begin{pmatrix} \frac{1}{7} \\ -\frac{3}{7} \\ 1 \end{pmatrix}$$

$$\underline{II} \cdot r_2 = 2$$

$$\begin{pmatrix} 1 & -1 & -1 \\ 3 & 1 & -3 \\ -4 & -2 & 1 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = r_2 \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$\Rightarrow \begin{cases} b_1 - b_2 - b_3 = 2b_1 \\ 3b_1 + b_2 - 3b_3 = 2b_2 \\ -4b_1 - 2b_2 + b_3 = 2b_3 \end{cases} \Rightarrow \begin{cases} -b_1 - b_2 - b_3 = 0 \\ 3b_1 - b_2 - 3b_3 = 0 \\ -4b_1 - 2b_2 - b_3 = 0 \end{cases}$$

$$\beta = \begin{vmatrix} -1 & -1 \\ 3 & -1 \end{vmatrix} = 1 + 3 = 4 \neq 0$$

$$\text{mit: } b_3 = \beta$$

$$\begin{cases} -b_1 - b_2 = \beta \\ 3b_1 - b_2 = 3\beta \quad (-) \end{cases}$$

$$\{ 3b_1 - b_2 = \frac{1}{3} \beta (-) \}$$

$$-4b_1 = -2\beta \Rightarrow b_1 = \frac{\beta}{2}$$

$$b_2 = -\frac{\beta}{2} - \frac{2}{\beta} = -\frac{3}{2}\beta$$

$$\Rightarrow \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \beta \begin{pmatrix} \frac{1}{2} \\ -\frac{3}{2} \\ 1 \end{pmatrix}$$

$$\underline{\text{III}}. \quad \pi_3 = -2$$

$$\begin{pmatrix} 1 & -1 & -1 \\ 3 & 1 & -3 \\ -4 & -2 & 1 \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \pi_3 \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$\begin{cases} c_1 - c_2 - c_3 = -2c_1 \\ 3c_1 + c_2 - 3c_3 = -2c_2 \\ -4c_1 - 2c_2 + c_3 = -2c_3 \end{cases} \Rightarrow \begin{cases} 3c_1 - c_2 - c_3 = 0 \\ 3c_1 + 3c_2 - 3c_3 = 0 \\ -4c_1 - 2c_2 + 3c_3 = 0 \end{cases}$$

$$\beta = \begin{vmatrix} 3 & -1 \\ 3 & 3 \end{vmatrix} = 9 + 3 = 12 \neq 0$$

$$\text{not: } c_3 = \varphi$$

$$\Rightarrow \begin{cases} 3C_1 - C_2 = \varphi \\ 3C_1 + 3C_2 = 3\varphi \quad (-) \end{cases}$$

$$-4C_2 = -2\varphi \Rightarrow C_2 = \frac{\varphi}{2}$$

$$\Rightarrow C_1 = \frac{\varphi + \frac{\varphi}{2}}{3} = \frac{\frac{3\varphi}{2}}{3} = \frac{\varphi}{2}$$

$$\Rightarrow \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = \varphi \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = C_1 \cdot e^{3t} \cdot \begin{pmatrix} \frac{1}{7} \\ -\frac{9}{7} \\ 1 \end{pmatrix} + C_2 \cdot e^{2t} \cdot \begin{pmatrix} \frac{1}{2} \\ -\frac{3}{2} \\ 1 \end{pmatrix} + C_3 \cdot e^{-2t} \cdot \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{pmatrix}$$

2) Ecuații cu derivate parțiale de ordinul I

1) $y^2 \frac{\partial u}{\partial x} + x^2 \frac{\partial u}{\partial y} + xy \frac{\partial u}{\partial z} = 0$

$$y^2 \frac{\partial u}{\partial x} + x^2 \frac{\partial u}{\partial y} + xy \frac{\partial u}{\partial z} = 0$$

$$\frac{dx}{y^2} = \frac{dy}{xz} = \frac{dz}{xy}$$

$$\frac{dx}{y^2} = \frac{dy}{xz} \quad | \cdot z \Rightarrow \frac{dx}{y} = \frac{dy}{x} \quad | \cdot (x \cdot y)$$

$$\Rightarrow x dx = y dy \quad | \int \Rightarrow \frac{x^2}{2} = \frac{y^2}{2} + C_1$$

$$\Rightarrow \frac{x^2}{2} - \frac{y^2}{2} = C_1 \rightarrow \text{integrală primă}$$

$$\frac{dy}{xz} = \frac{dz}{xy} \quad | \cdot (x y z) \Rightarrow$$

$$\Rightarrow y dy = z dz \quad | \int \Rightarrow \frac{y^2}{2} = \frac{z^2}{2} + C_2$$

$$\Rightarrow \frac{y^2}{2} - \frac{z^2}{2} = C_2 \rightarrow \text{integrală primă}$$

$$\Rightarrow \begin{cases} \frac{x^2}{2} - \frac{y^2}{2} = C_1 \Leftrightarrow x^2 - y^2 = C_1 \\ \frac{y^2}{2} - \frac{z^2}{2} = C_2 \Leftrightarrow y^2 - z^2 = C_2 \end{cases}$$

$$\Rightarrow \phi(x^2 - y^2, y^2 - z^2) = 0$$

$$2) \quad 2) \quad xz \frac{\partial z}{\partial x} + yz \frac{\partial z}{\partial y} = z^2$$

$$\frac{dx}{xz} = \frac{dy}{yz} = \frac{dz}{z^2}$$

$$\frac{dx}{xz} = \frac{dy}{yz} \quad | \cdot z \Rightarrow \frac{dx}{x} = \frac{dy}{y} \Rightarrow$$

$$\Rightarrow \ln x = \ln y + \ln C_1 \Rightarrow \ln x = \ln (y \cdot C_1)$$

$$\Rightarrow C_1 = \left[\frac{x}{y} \right]$$

$$\frac{dy}{yz} = \frac{dz}{z^2} \quad | \cdot z \Rightarrow \frac{dy}{y} = \frac{dz}{z} \Rightarrow$$

$$\Rightarrow \ln y = \ln z + \ln C_2 \Rightarrow C_2 = \left[\frac{y}{z} \right]$$

$$\Rightarrow \phi\left(\frac{x}{y}, \frac{y}{z}\right) = 0$$

$$3) \quad x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + (x+y) \frac{\partial u}{\partial z} = 0$$

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{x+y}$$

$$\frac{dx}{x} = \frac{dy}{y} \Rightarrow \ln x = \ln y + \ln C_1 \Rightarrow C_1 = \frac{x}{y}$$

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{x+y}$$

$$\frac{dx + dy}{x+y} = \frac{dz}{x+y} \quad | \cdot (x+y)$$

$$\Rightarrow d(x+y) = dz \Rightarrow x+y = z + C_2$$

$$\Rightarrow C_2 = x+y-z$$

$$\Rightarrow \phi\left(\frac{x}{y}, x+y-z\right)$$

$$4) \quad (y+z) \frac{\partial u}{\partial x} + (z+x) \frac{\partial u}{\partial y} = x+y$$

$$\underline{dx} = dy - dz$$

$$\frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y}$$

$$\frac{dx - dy}{-(x-y)} = \frac{dy - dz}{-(y-z)} \quad (\Rightarrow) \quad \frac{d(x-y)}{-(x-y)} = \frac{d(y-z)}{-(y-z)} \quad \int \Rightarrow$$

$$\Rightarrow -\ln(x-y) = -\ln(y-z) - \ln C_1$$

$$\Rightarrow C_1 = \frac{x-y}{y-z}$$

$$\frac{dx - dz}{-(x-z)} = \frac{dx + dy + dz}{2(x+y+z)} \quad \int \Rightarrow$$

$$-\ln(x-z) = \frac{1}{2} \ln(x+y+z) + \frac{1}{2} \ln C_2 \quad | \cdot 2$$

$$\Rightarrow (x-z)^2 = (x+y+z) \cdot C_2$$

$$\Rightarrow C_2 = \frac{(x-z)^2}{(x+y+z)} \quad \int^{-1}$$

$$C_2 = (x-z)^2 \cdot (x+y+z)$$

$$\Rightarrow \phi \left(\frac{x-y}{y-z}, (x-z)^2 (x+y+z) \right)$$

$$5) \quad x(y^2 - z^2) \frac{\partial z}{\partial x} + y(z^2 - x^2) \frac{\partial z}{\partial y} = z(x^2 - y^2)$$

$$\frac{dx}{x(y^2 - z^2)} = \frac{dy}{y(z^2 - x^2)} = \frac{dz}{z(x^2 - y^2)}$$

$$\frac{\frac{dx}{x}}{y^2 - z^2} = \frac{\frac{dy}{y}}{z^2 - x^2} = \frac{\frac{dz}{z}}{x^2 - y^2}$$

$$\frac{\frac{dx}{x} + \frac{dy}{y}}{-(x^2 - y^2)} = \frac{\frac{dz}{z}}{x^2 - y^2} \quad | \cdot (- (x^2 - y^2))$$

$$\frac{dx}{x} + \frac{dy}{y} = -\frac{dz}{z} \quad \int$$

$$\ln x + \ln y + \ln z = \ln C_1$$

$$\Rightarrow C_1 = x \cdot y \cdot z$$

$$\frac{x}{x(y^2 - z^2)} = \frac{y}{y(z^2 - x^2)} = \frac{z}{z(x^2 - y^2)}$$

$$\frac{x dx}{x^2 y^2 - x^2 z^2} = \frac{y dy}{y^2 z^2 - y^2 x^2} = \frac{z dz}{z^2 x^2 - z^2 y^2} = \frac{x dx + y dy + z dz}{0}$$

$$\Rightarrow x dx + y dy + z dz = 0 \quad \int \Rightarrow$$

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{C_2}{2}$$

$$\Rightarrow C_2 = x^2 + y^2 + z^2$$

$$\Rightarrow \phi(x, y, z, x^2 + y^2 + z^2)$$

$$6) \quad (z^2 - y^2) \frac{\partial u}{\partial x} + z \frac{\partial u}{\partial y} - y \frac{\partial u}{\partial z} = 0$$

$$\frac{dx}{z^2 - y^2} = \frac{dy}{z} = \frac{dz}{-y}$$

$$\frac{dx}{z^2 - y^2} = \frac{z dx + y dz}{z^2 - y^2} \quad | \cdot (z^2 - y^2)$$

$$\Rightarrow dx = z dx + y dz \quad \int \Rightarrow$$

$$x = xz + yz + C_1$$

$$\Rightarrow C_1 = x - z(x+y)$$

$$\frac{dx}{z^2 - y^2} = \frac{dy}{z} = \frac{dz}{-y}$$

$$\frac{dx}{(\cancel{z-y})(z+y)} = \frac{d(\cancel{z+y})}{(\cancel{z-y})} \mid \cdot (z+y)$$

$$dx = d(z+y) \cdot (z+y) \mid \int \Rightarrow$$

$$\Rightarrow x = \frac{(z+y)^2}{2} + \frac{C_2}{2} \mid \cdot 2$$

$$\Rightarrow C_2 = 2x - (z+y)^2$$

$$\Rightarrow \phi(x - z(x+y), 2x - (z+y)^2)$$

7)

$$\begin{cases} \frac{\partial}{\partial x} \sqrt{x} \frac{\partial z}{\partial x} + \sqrt{y} \frac{\partial z}{\partial y} = \sqrt{z} \\ y = z \\ u(x, y, z) = x - y \end{cases}$$

$$\frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}}$$

$$\frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}} \xrightarrow{\int} \int x^{-\frac{1}{2}} dx = \int y^{-\frac{1}{2}} dy$$

$$\Rightarrow \frac{x^{\frac{1}{2}}}{\frac{1}{2}} = \frac{y^{\frac{1}{2}}}{\frac{1}{2}} + C_1 \Rightarrow C_1 = \sqrt{x} - \sqrt{y}$$

$$\frac{dx}{\sqrt{x}} = \frac{dz}{\sqrt{z}} \xrightarrow{\int} \int \frac{1}{\sqrt{x}} dx = \int \frac{1}{\sqrt{z}} dz \Rightarrow C_2 = \sqrt{x} - \sqrt{z}$$

$$\begin{cases} \sqrt{x} - \sqrt{y} = C_1 \\ \sqrt{x} - \sqrt{z} = C_2 \Rightarrow \sqrt{x} - 1 = C_2 \Rightarrow x = (C_2 + 1)^2 \\ z = 1 \\ U(x, y, z) = x - y \end{cases}$$

$$\Rightarrow C_2 + 1 - \sqrt{y} = C_1 \Rightarrow y = (C_2 - C_1 + 1)^2$$

$$\Rightarrow U(x, y, z) = (C_2 + 1)^2 - (C_2 - C_1 + 1)^2$$

8)

$$\begin{cases} (x^2 + y^2) \frac{\partial z}{\partial x} + 2xy \frac{\partial z}{\partial y} = xz \\ z = 1 \\ x^2 - y^2 = 1 \end{cases}$$

$$\frac{dx}{x^2 + y^2} = \frac{dy}{2xy} = \frac{dz}{xz}$$

$$\frac{dx}{x^2+y^2} = \frac{dy}{2xy} \Rightarrow \frac{dx+dy}{x^2+2xy+y^2} = \frac{dx-dy}{x^2-2xy+y^2} \int \rightarrow$$

$$\Rightarrow \int \frac{1}{(x+y)^2} d(x+y) = \int \frac{1}{(x-y)^2} d(x-y) \Rightarrow -\frac{1}{x+y} = -\frac{1}{x-y} + C_1$$

$$\Rightarrow C_1 = \frac{\overset{x+y}{1}}{x-y} - \frac{\overset{x-y}{1}}{x+y} = \frac{x+y - x+y}{x^2-y^2} = \frac{2y}{x^2-y^2}$$

$$\frac{dy}{2xy} = \frac{dz}{xz} \quad | \cdot x \Rightarrow \frac{dy}{2y} = \frac{dz}{z} \int \Rightarrow$$

$$\Rightarrow \frac{1}{2} \ln y = \ln z + \ln C \quad | \cdot 2$$

$$y = z^2 \cdot C_2 \Rightarrow C_2 = \frac{y}{z^2}$$

$$\left\{ \begin{array}{l} C_1 = \frac{2y}{x^2-y^2} \Rightarrow C_1 = 2y \\ C_2 = \frac{y}{z^2} \Rightarrow C_2 = y \\ z = 1 \\ x^2 - y^2 = 1 \end{array} \right. \quad \begin{array}{l} \Rightarrow C_1 = 2C_2 \Rightarrow \\ \frac{C_1}{C_2} = 2 \end{array}$$

$$\frac{2y}{x^2-y^2}$$

n 1

n 2

n 3

$$\frac{\frac{2y}{x^2-y^2}}{\frac{y}{z^2}} = 2 \quad (\Rightarrow) \quad \frac{2y}{x^2-y^2} \cdot \frac{z^2}{y} = 2 \quad (\Rightarrow) \quad \frac{z^2}{x^2-y^2} = 1$$

g)

$$\begin{cases} (z-y)^2 \frac{\partial u}{\partial x} + x \frac{\partial u}{\partial y} + y \frac{\partial u}{\partial z} = 0 \\ u(0, y, z) = 2y(y-z) \end{cases}$$

$$\frac{dx}{(z-y)^2} = \frac{dy}{z} = \frac{dz}{y}$$

$$\frac{dx}{(z-y)^2} = \frac{dy - dz}{z - y} \quad (\Rightarrow) \quad \frac{dx}{z-y} = \frac{d(y-z)}{1}$$

$$\Rightarrow dx = (z-y) d(y-z) \quad \int \Rightarrow x = \frac{(y-z)^2}{2} + \frac{C_1}{2}$$

$$\Rightarrow C_1 = 2x + (z-y)^2$$

$$\frac{dy}{z} = \frac{dz}{y} \quad | \cdot (zy) \Rightarrow y dy = z dz \quad \int \Rightarrow$$

$$\frac{y^2}{2} = \frac{z^2}{2} + \frac{C_2}{2} \Rightarrow C_2 = y^2 - z^2$$

$$\begin{cases} C_1 = 2x + (z-y)^2 \Rightarrow 2x + z^2 + y^2 - 2yz = C_1 \\ C_2 = y^2 - z^2 \Rightarrow y^2 = C_2 + z^2 \end{cases}$$

$$\begin{cases} C_2 = y^2 - z^2 \Rightarrow y^2 = C_2 + z^2 \\ U(0, y, z) = 2y(y - z) \end{cases}$$

$$\Rightarrow 2yz = 2x + z^2 + y^2 - C_1$$

$$U(0, y, z) = 2y(y - z) \quad , \quad x = 0 \Rightarrow$$

$$\Rightarrow 2yz = z^2 + y^2 + C_1$$

$$y^2 = C_2 + z^2$$

$$U = 2(C_2 + z^2) - z^2 - y^2 - C_1$$

$$= \cancel{2y^2} - \cancel{2z^2} + \cancel{2z^2} - z^2 - \cancel{y^2} - 2x - z^2 - \cancel{y^2} + 2yz$$

$$= -2z^2 - 2x + 2yz$$

10)
$$\begin{cases} xz \frac{\partial z}{\partial x} - yz \frac{\partial z}{\partial y} = x^2 + y^2 \\ z = 1 \\ x^2 - y^2 - 2xy = 1 \end{cases}$$

$$\frac{dx}{xz} = \frac{dy}{-yz} = \frac{dz}{x^2 + y^2}$$

$$\frac{dx}{xz} = \frac{dy}{-yz} \Big| \cdot z \Rightarrow \frac{dx}{x} = \frac{dy}{-y} \quad \int \Rightarrow$$

$$\frac{u}{x^2} = \frac{-y}{-y^2} \cdot z \Rightarrow \frac{u}{x} = \frac{-y}{-y} \Rightarrow$$

$$\Rightarrow \ln x = -\ln y + \ln C_1 \Rightarrow C_1 = x \cdot y$$

$$\frac{dx}{x^2} = \frac{dy}{-y^2} = \frac{dz}{x^2+y^2} = \frac{-x dx + y dz + z dz}{-\cancel{x^2} - \cancel{y^2} + \cancel{z} x^2 + \cancel{z} y^2}$$

$$\Rightarrow -x dx + y dy + z dz = 0 \quad \int$$

$$-\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = C_2 \Rightarrow -x^2 + y^2 + z^2 = C_2$$

$$\begin{cases} C_1 = x y \\ C_2 = -x^2 + y^2 + z^2 \rightarrow C_2 = -x^2 + y^2 + 1 \\ z = 1 \\ x^2 - y^2 - 2xy = 1 \end{cases}$$

$$\begin{aligned} & z = -x^2 + y^2 + x^2 - y^2 - 2xy \\ & = -2xy \end{aligned}$$

$$C_2 = -2xy = -2 C_1 \Rightarrow \frac{C_2}{C_1} = -2$$

$$\frac{-x^2 + y^2 + z^2}{xy} = -2$$

3) • Să se calculeze derivata următoarelor câmpuri scalare după direcția vectorului $\vec{v}$

1) 1) $f(x, y, z) = x^2 y^3 z^4$, $\vec{v} = 2\vec{i} + 2\vec{j} - \vec{k}$

$$f(x, y, z) = x^2 y^3 z^4 \quad \vec{v} = 2\vec{i} + 2\vec{j} - \vec{k}$$

$$|\vec{v}| = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$$

$$\begin{cases} \alpha = \frac{2}{3} \\ \beta = \frac{2}{3} \\ \gamma = -\frac{1}{3} \end{cases}$$

$$\Rightarrow \frac{df}{d\vec{v}} = \alpha \cdot \frac{\partial f}{\partial x} + \beta \cdot \frac{\partial f}{\partial y} + \gamma \cdot \frac{\partial f}{\partial z}$$

$$\frac{df}{d\vec{v}} = \frac{2}{3} \cdot 2 \cdot x y^3 z^4 + \frac{2}{3} \cdot 3 x^2 y^2 z^4 - \frac{1}{3} \cdot 4 x^2 y^3 z^3$$

$$\frac{df}{d\vec{v}} = \frac{4}{3} x y^3 z^4 + 2 x^2 y^2 z^4 - \frac{4}{3} x^2 y^3 z^3$$

2) 2) $f(x, y, z) = \sin(xy z)$, $\vec{v} = \vec{i} - \vec{j} - \vec{k}$

$$f(x, y, z) = \sin(xy z) \quad \vec{v} = \vec{i} - \vec{j} - \vec{k}$$

$$|\vec{v}| = \sqrt{1+1+1} = \sqrt{3}$$

$$\alpha = \frac{1}{\sqrt{3}}$$

$$\beta = -\frac{1}{\sqrt{3}}$$

$$\gamma = -\frac{1}{\sqrt{3}}$$

$$\left| \begin{array}{l} \frac{\partial f}{\partial x} = \cos(xyz) \cdot yz \\ \frac{\partial f}{\partial y} = \cos(xyz) \cdot xz \\ \frac{\partial f}{\partial z} = \cos(xyz) \cdot xy \end{array} \right.$$

$$\frac{df}{d\vec{v}} = \alpha \cdot \frac{\partial f}{\partial x} + \beta \frac{\partial f}{\partial y} + \gamma \frac{\partial f}{\partial z}$$

$$\frac{df}{d\vec{v}} = \frac{\cos(xyz)}{\sqrt{3}} (yz - xz - xy)$$

3) 3) $f(x, y, z) = \frac{xy}{z}$, $\vec{v} = 3\vec{i} - 5\vec{k}$

$$f(x, y, z) = \frac{xy}{z} \quad \vec{v} = 3\vec{i} - 5\vec{k}$$

$$|\vec{v}| = \sqrt{9+25} = \sqrt{34}$$

$$\alpha = \frac{3}{\sqrt{34}}$$

$$\beta = 0$$

$$\gamma = -\frac{5}{\sqrt{34}}$$

$$\left| \begin{array}{l} \frac{\partial f}{\partial x} = \frac{y}{z} \\ \frac{\partial f}{\partial y} = \frac{x}{z} \\ \frac{\partial f}{\partial z} = -\frac{xy}{z^2} \end{array} \right.$$

$$\begin{aligned}\frac{df}{d\vec{v}} &= \alpha \frac{\partial f}{\partial x} + \beta \cdot \frac{\partial f}{\partial y} + \gamma \frac{\partial f}{\partial z} \\ &= \frac{1}{\sqrt{34}} \left(3 \frac{y}{z} + 5 \frac{xy}{z^2} \right)\end{aligned}$$

4) • Să se calculeze gradientul următoarelor câmpuri scalare

1) $f(x, y, z) = e^{xy^2} + \sin(x + 2y + 3z)$

$$f(x, y, z) = e^{xy^2} + \sin(x + 2y + 3z)$$

$$\text{grad } f = \nabla \cdot f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

$$\frac{\partial f}{\partial x} = y^2 e^{xy^2} + \cos(x + 2y + 3z)$$

$$\frac{\partial f}{\partial y} = xz e^{xy^2} + \cos(x + 2y + 3z) \cdot 2$$

$$\frac{\partial f}{\partial z} = xy e^{xy^2} + \cos(x + 2y + 3z) \cdot 3$$

$$\begin{aligned}\Rightarrow \text{grad } f &= [y^2 e^{xy^2} + \cos(x + 2y + 3z)] \vec{i} + [xz e^{xy^2} + \\ &+ 2 \cos(x + 2y + 3z)] \vec{j} + [xy e^{xy^2} + \cos(x + 2y + 3z) \cdot 3] \vec{k}\end{aligned}$$

$$2) \ 2) f(x, y, z) = (2x + 3y) e^{xy^2} - \ln(x^2 + y + 3z)$$

$$f(x, y, z) = (2x + 3y) e^{xy^2} - \ln(x^2 + y + 3z)$$

$$\frac{\partial f}{\partial x} = 2 e^{xy^2} + (2x + 3y) \cdot y^2 \cdot e^{xy^2} - \frac{1}{x^2 + y + 3z} \cdot 2x$$

$$\frac{\partial f}{\partial y} = 3 e^{xy^2} + (2x + 3y) \cdot x^2 \cdot e^{xy^2} - \frac{1}{x^2 + y + 3z}$$

$$\frac{\partial f}{\partial z} = (2x + 3y) \cdot xy \cdot e^{xy^2} - \frac{1}{x^2 + y + 3z} \cdot 3$$

$$\text{grad } f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

$$= \left[2 e^{xy^2} + (2x + 3y) \cdot y^2 \cdot e^{xy^2} - \frac{1}{x^2 + y + 3z} \cdot 2x \right] \cdot \vec{i}$$

$$+ \left[3 e^{xy^2} + (2x + 3y) \cdot x^2 \cdot e^{xy^2} - \frac{1}{x^2 + y + 3z} \right] \cdot \vec{j}$$

$$+ \left[(2x + 3y) \cdot xy \cdot e^{xy^2} - \frac{1}{x^2 + y + 3z} \cdot 3 \right] \cdot \vec{k}$$

$$3) \ 3) f(x, y, z) = \arctg(x + 2y + 3z)$$

$$f(x, y, z) = \arctg(x + 2y + 3z)$$

$$\frac{\partial f}{\partial x} = \frac{1}{1 + (x + 2y + 3z)^2}$$

$$\frac{\partial f}{\partial y} = \frac{1}{1 + (x + 2y + 3z)^2} \cdot 2$$

$$\frac{\partial f}{\partial z} = \frac{1}{1 + (x + 2y + 3z)^2} \cdot 3$$

$$\text{grad } f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

$$= \frac{1}{1 + (x + 2y + 3z)^2} \cdot \vec{i} + \frac{2}{1 + (x + 2y + 3z)^2} \cdot \vec{j} + \frac{3}{1 + (x + 2y + 3z)^2} \cdot \vec{k}$$

5) • Să se calculeze divergența și rotorul următoarelor câmpuri vectoriale

1) $\vec{v} = x^2 e^{yz} \vec{i} + [y^2 + \ln(xz)] \vec{j} + \frac{z^2}{x + 2z^3} \vec{k}$

$$\vec{v} = x^2 e^{yz} \vec{i} + [y^2 + \ln(xz)] \vec{j} + \frac{z^2}{x + 2z^3} \vec{k}$$

$$\operatorname{div} \vec{v} = \nabla \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$= 2x e^{y^2} + 2y + \frac{2z(x + 2z^3) - z^2 \cdot 6z^2}{(x + 2z^3)^2}$$

$$\operatorname{rot} \vec{v} = \nabla \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} =$$

$$= \vec{i} \left[\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right] + \vec{j} \left[\frac{\partial v_1}{\partial z} - \frac{\partial v_3}{\partial x} \right] + \vec{k} \left[\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right]$$

$$= \vec{i} \left[0 - \frac{1}{x^2} \cdot x \right] + \vec{j} \left[x^2 y e^{y^2} + \frac{z^2}{(x + 2z^3)^2} \right] + \vec{k} \left[\frac{1}{x^2} \cdot x - x^2 z e^{y^2} \right]$$

$$= -\frac{1}{x} \vec{i} + \left[x^2 y e^{y^2} + \frac{z^2}{(x + 2z^3)^2} \right] \vec{j} + \left[\frac{1}{x} - x^2 z e^{y^2} \right] \vec{k}$$

$$2) \quad \vec{v} = y \cos(xz^2) \vec{i} - y^2 \cos(xz) \vec{j} + x \sin(y^2 z) \vec{k}$$

$$\vec{v} = y \cos(xz^2) \vec{i} - y^2 \cos(xz) \vec{j} + x \sin(y^2 z) \vec{k}$$

$$\operatorname{div} \vec{v} = \nabla \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$= -y \sin(xz^2) \cdot z^2 - 2y \cos(xz) + x \cos(y^2 z) \cdot y^2$$

$$= -y \sin(xz^2) \cdot z^2 - 2y \cos(xz^2) + x \cos(yz^2) \cdot y$$

$$\text{rot } \vec{v} = \nabla \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} =$$

$$= \vec{i} \left[\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right] + \vec{j} \left[\frac{\partial v_1}{\partial z} - \frac{\partial v_3}{\partial x} \right] + \vec{k} \left[\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right]$$

$$= \vec{i} \left[x \cos(y^2 z) \cdot 2yz - y^2 \sin(xz^2) \cdot x \right] + \vec{j} \left[-y \sin(xz^2) \cdot 2xz - \sin(y^2 z) \right] + \vec{k} \left[y^2 \sin(xz^2) \cdot z - \cos(xz^2) \right]$$

$$3) \vec{v} = 6xy \vec{i} + (x^2 + \ln y) \vec{j} + z^2 \ln z \vec{k}$$

$$\vec{v} = 6xy \vec{i} + (x^2 + \ln y) \vec{j} + z^2 \ln z \vec{k}$$

$$\text{div } \vec{v} = \nabla \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$= 6y + \frac{1}{y} + 2z \ln z + z^2 \cdot \frac{1}{z}$$

$$= 6y + \frac{1}{y} + 2z \ln z + z$$

$$\text{rot } \vec{v} = \nabla \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix} =$$

$$\text{rot } \vec{v} = \nabla \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} =$$

$$= \vec{i} \left[\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right] + \vec{j} \left[\frac{\partial v_1}{\partial z} - \frac{\partial v_3}{\partial x} \right] + \vec{k} \left[\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right]$$

$$= \vec{i} [0 - 0] + \vec{j} [0 - 0] + \vec{k} [2x - 6x]$$

$$= -4x \vec{k}$$

4) $\vec{v} = (e^x - xz)\vec{i} + (ze^{xy} - e^{zy})\vec{j} - xyz^4\vec{k}$

$$\vec{v} = (e^x - xz)\vec{i} + (ze^{xy} - e^{zy})\vec{j} - xyz^4\vec{k}$$

$$\text{div } \vec{v} = \nabla \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$= e^x - z + z \times e^{xy} - 2e^{zy} - 4xyz^3$$

$$\text{rot } \vec{v} = \nabla \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} =$$

$$= \vec{i} \left[\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right] + \vec{j} \left[\frac{\partial v_1}{\partial z} - \frac{\partial v_3}{\partial x} \right] + \vec{k} \left[\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right]$$

$$= \vec{i} \left[\frac{\partial V_3}{\partial y} - \frac{\partial V_2}{\partial z} \right] + \vec{j} \left[\frac{\partial V_1}{\partial z} - \frac{\partial V_3}{\partial x} \right] + \vec{k} \left[\frac{\partial V_2}{\partial x} - \frac{\partial V_1}{\partial y} \right]$$

$$= \vec{i} \left[-xz^4 - e^{xy} \right] + \vec{j} \left[-x + yz^4 \right] + \vec{k} \cdot zy e^{xy}$$